# Lecture 7: Planetary Surfaces
# Geology, Geomorphology, & Geophysics

## Introduction to Planetary Science


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Olympus Mons, Mars. Credit: NASA/JPL/USGS

# Learning Objectives

By the end of this lecture, you will be able to:

- Distinguish endogenic and exogenic surface processes
- Describe the three stages of impact crater formation
- Derive the crater scaling law $D \sim (E/\rho g)^{1/4}$ from dimensional analysis
- Use crater counting to determine relative and absolute surface ages
- Compare volcanic, tectonic, and erosional styles across the solar system
- Explain cryovolcanism on icy ocean worlds

# Recap: Lectures 5 & 6

- Atmospheric structure: hydrostatic equilibrium, scale height, vertical layers
- Energy balance and greenhouse effect: $T_s > T_{\mathrm{eff}}$ from IR absorption
- Clausius-Clapeyron equation: cloud formation across the solar system
- Atmospheric dynamics: Hadley cells, Coriolis effect, jet streams
- Carbonate-silicate cycle: Earth's long-term climate thermostat
- Faint young Sun paradox; Venus runaway greenhouse; Mars climate puzzle

**Today:** What shapes the *surfaces* of planets and moons?

# 1. Surface Processes

Mars surface, Curiosity rover. Credit: NASA/JPL-Caltech/MSSS

# Endogenic vs. Exogenic Processes

A planet's surface is its geological record: billions of years of competing processes that create, modify, and destroy landforms.

**Endogenic** (internal heat):

- Volcanism
- Tectonics (faulting, folding)
- More vigorous for larger, hotter bodies

**Exogenic** (external forces):

- Impact cratering
- Erosion (wind, water, ice)
- Space weathering, radiation

The balance between these processes determines what we see on a surface.

# Dominant Surface Processes by Body

| Body | Dominant process | Key evidence |
|---|---|---|
| Moon | Impact cratering | Saturated highland craters, >4 Gyr |
| Earth | Plate tectonics + erosion | Ocean floors <200 Myr; rapid weathering |
| Mars | Volcanism + impacts | Olympus Mons; cratered southern highlands |
| Venus | Volcanism | Volcanic plains cover ~80% of surface |
| Io | Tidal volcanism | ~400 active volcanoes; age <1 Myr |
| Europa | Ice tectonics | Lineae, chaos terrain, very few craters |

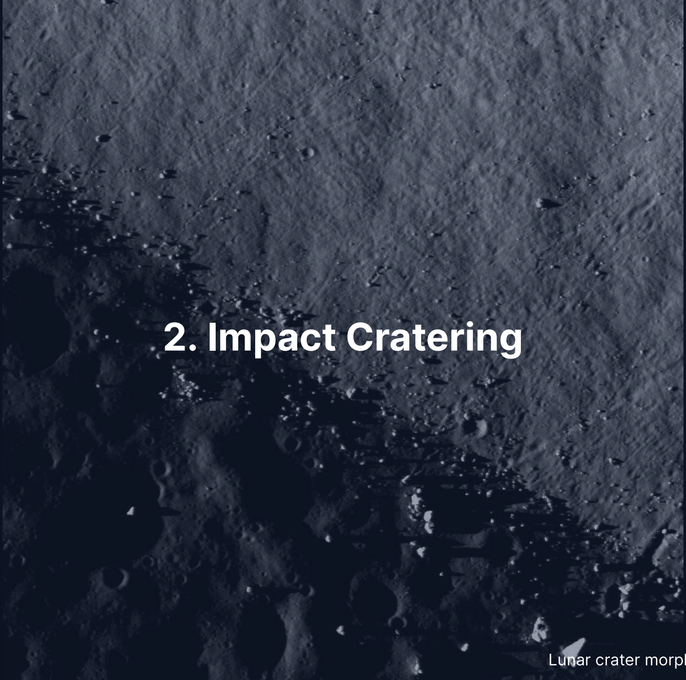
Internal heat → endogenic processes dominate.
Dead interior → surface preserves ancient craters.

# Reading the Surface Record

Credit: NASA/JPL-Caltech/MSSS

- Aeolian dunes sculpted by wind in Gale Crater, Mars
- Wind erosion dominates Mars today
- Ancient volcanism and impacts shaped the deeper record
- On Earth, erosion destroys craters within ~10 Myr

# 2. Impact Cratering



Lunar crater morphology. Credit: NASA/GSFC/ASU

# The Most Universal Geological Process

- **Every** solid body in the solar system bears impact craters
- On bodies without atmospheres or geology (Moon, Mercury), craters survive billions of years
- Impact velocities: 10–70 $\mathrm{km\,s^{-1}}$ (set by orbital mechanics)
- Enormous energy release in a fraction of a second

Three stages of crater formation:

1. Contact and compression
2. Excavation
3. Modification

## Stage 1: Contact and Compression

- Projectile strikes at 10–70 km $s^{-1}$
- Shock wave propagates into both target and impactor
- Peak pressures ~100 GPa (comparable to Earth's core–mantle boundary)
- Material near impact point is **vaporised and melted**
- Kinetic energy → internal energy in < 1 s

The kinetic energy of an impactor:

$$E_k = \frac{1}{2}mv^2$$

# Example: 1 km Asteroid Impact

Rocky asteroid: $\rho \approx 3000\ \mathrm{kg\,m^{-3}}$, diameter = 1 km

$$m = \frac{4}{3}\pi r^3 \rho \approx \frac{4}{3}\pi (500)^3 \times 3000 \approx 1.6 \times 10^{12}\ \mathrm{kg}$$

At $v = 20\ \mathrm{km\,s^{-1}}$:

$$E_k = \frac{1}{2} \times 1.6 \times 10^{12} \times (2 \times 10^4)^2 \approx 3 \times 10^{20}\ \mathrm{J}$$

~100× the energy of the largest nuclear weapon ever detonated, released in <1 s at a single point.

# Stages 2 & 3: Excavation and Modification

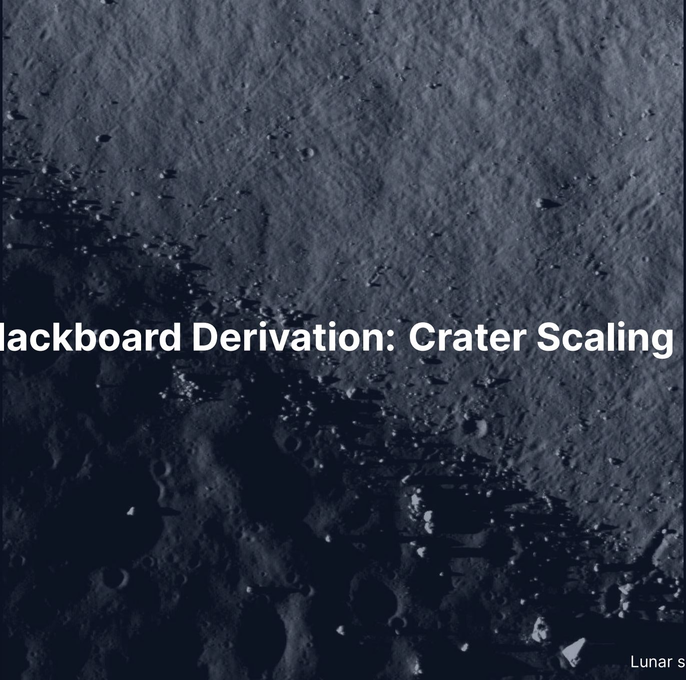
**Excavation:**

- Shock wave expands hemispherically outward
- Rarefaction wave accelerates material upward and outward
- **Transient cavity** forms (depth:diameter ≈ 1:3)
- Cone-shaped **ejecta blanket** deposited around the crater

**Modification:**

- Transient cavity is gravitationally unstable
- Small craters: walls slump inward → **simple crater** (bowl-shaped)
- Large craters: floor rebounds, walls collapse in terraces → **complex crater** with central peak

# 3. Blackboard Derivation: Crater Scaling Law



Lunar surface. Credit: NASA/GSFC/ASU

# Setup: Crater Size from Dimensional Analysis

**Goal:** How does crater diameter $D$ depend on impact energy $E$, target density $\rho$, and surface gravity $g$?

In the **gravity regime** (craters ≳100 m, where gravity limits size):

$$D = C\, E^a \rho^b g^c$$

Dimensions ($M$, $L$, $T$):

$$\begin{aligned}
[D] &= L \\
[E] &= ML^2T^{-2} \\
[\rho] &= ML^{-3} \\
[g] &= LT^{-2}
\end{aligned}$$

# Solving for the Exponents

Require $[E^a \rho^b g^c] = L$:

**Mass:** $a + b = 0 \implies b = -a$

**Time:** $-2a - 2c = 0 \implies c = -a$

**Length:** $2a - 3b + c = 1$

Substituting $b = -a$ and $c = -a$:

$$2a + 3a - a = 1 \implies 4a = 1 \implies a = \frac{1}{4}$$

Therefore $a = 1/4$, $b = -1/4$, $c = -1/4$:

$$\boxed{D \sim \left(\frac{E}{\rho g}\right)^{1/4}}$$

# Interpretation of the Scaling Law

- *D* scales as the **fourth root** of energy
- Doubling the energy increases *D* by only $2^{1/4} \approx 1.19$ (19%)
- This explains the relatively narrow range of crater sizes even though impactor energies span many orders of magnitude

**Dimensional check:**

$$\left[\frac{E}{\rho g}\right] = \frac{ML^2T^{-2}}{ML^{-3} \cdot LT^{-2}} = L^4 \quad \Rightarrow \quad [D] = L$$

The more complete **pi-scaling framework** (Holsapple & Schmidt, 1993) parameterises the transition between the gravity and strength regimes.

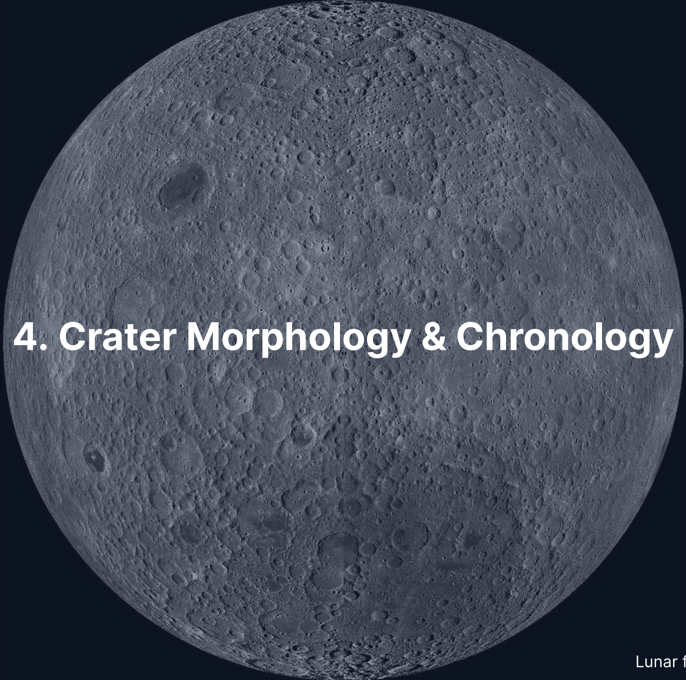
# Worked Example: 1 km Asteroid on the Moon

| Parameter | Symbol | Value |
|---|---|---|
| Impact energy | $E$ | $3 \times 10^{20}$ J |
| Regolith density | $\rho$ | 2500 kg m$^{-3}$ |
| Surface gravity | $g$ | 1.62 m s$^{-2}$ |

$$D \sim \left(\frac{3 \times 10^{20}}{2500 \times 1.62}\right)^{1/4} = (7.4 \times 10^{16})^{1/4} \approx 17\ \text{km}$$

Consistent with observed lunar craters from ~1 km impactors.
For comparison, the 85 km crater Tycho required a ~8 km impactor.

# 4. Crater Morphology & Chronology



Lunar farside. Credit: NASA/GSFC/ASU

# Crater Morphology: Three Classes

Credit: NASA/GSFC/ASU

1. **Simple** ($D < D_t$)
   Bowl-shaped; depth:diameter ≈ 1:5
   Moon: $D_t \approx 15$ km
2. **Complex** ($D_t < D \lesssim 300$ km)
   Central peak, terraced walls, flat floor
   Gravity overcomes wall strength
3. **Multi-ring basins** ($D \gtrsim 300$ km)
   Concentric ring structures
   Orientale (930 km), Caloris (1550 km), Hellas (2300 km)

## Transition Diameter

The simple-to-complex transition scales with surface gravity:

$$D_t \approx D_{t,\mathrm{Moon}} \cdot \frac{g_{\mathrm{Moon}}}{g}$$

| **Body** | $g$ **(m s$^{-2}$)** | $D_t$ **(km)** |
|---|---|---|
| Moon | 1.62 | 15 |
| Mars | 3.72 | ~6.5 |
| Earth | 9.81 | ~2.5 |
| Mercury | 3.70 | ~6.6 |

On Earth, virtually all craters larger than a few km are complex.

# Crater Size-Frequency Distribution

The number of craters larger than a given diameter $D$ follows an approximate power law:

$$N(>D) \propto D^{-b}, \quad b \approx 2\text{–}3$$

- Small craters are much more numerous than large ones
- Slope $b$ depends on impactor size distribution and erosion
- A plot of $\log N$ vs. $\log D$ yields a straight line for a pristine surface
- Higher cumulative density $\Rightarrow$ older surface
- Deviations from the power law indicate resurfacing, saturation, or secondary craters

Source: Neukum et al. (2001), *Space Sci. Rev.*

# Crater Counting: Principle

**Key idea:** Older surfaces have had more time to accumulate craters.

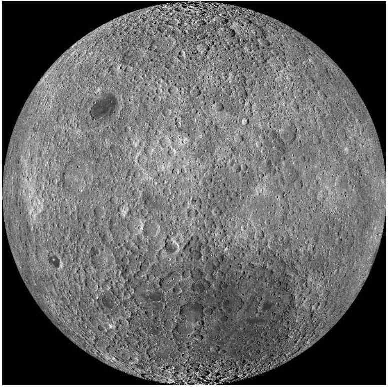
The **crater size-frequency distribution (SFD)** follows a power law:

$$N(> D) = a\, D^{-b}$$

- $N(> D)$: cumulative number of craters per unit area with diameter $> D$
- Production population: $b \approx 2–3$
- Plot $N$ vs. $D$ on log-log axes → straight line
- Higher line = older surface

Crater counting gives **relative ages**. Apollo/Luna samples provide **absolute calibration**.

# The Lunar Chronology



Credit: NASA/GSFC/ASU

Apollo and Luna samples give radiometric ages for specific surfaces:

- **Highlands:** saturated, >4 Gyr
- **Maria:** 3.9–3.1 Gyr
- Calibrated SFD → absolute ages

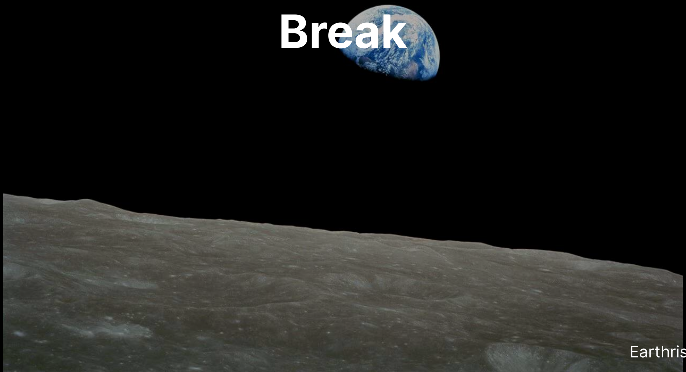
Extrapolated to other bodies:

- Mars southern highlands: ~4 Gyr
- Mars northern lowlands: much younger
- Venus: uniform crater density → ~300–700 Myr

# Key Results from Crater Chronology

- **Late Heavy Bombardment** (~3.9 Ga): spike in impact rate recorded on the Moon
- **Mars hemispheric dichotomy:**
  Southern highlands heavily cratered (~4 Gyr)
  Northern lowlands much smoother → volcanic resurfacing
- **Venus:** Remarkably uniform crater density
  Mean surface age ~300–700 Myr
  Implies a **global resurfacing event**
- **Europa:** Very few craters → surface age ~40–90 Myr
  Continuously resurfaced by ice tectonics

# Break



Earthrise, Apollo 8 (1968). Credit: NASA

# 5. Volcanism

Olympus Mons, Mars. Credit: NASA/JPL/USGS

# Effusive vs. Explosive Volcanism

The key variable is **magma viscosity**, controlled by $\mathrm{SiO_2}$ content:

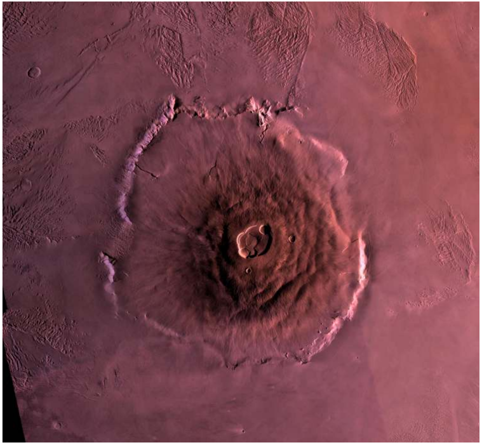
**Low viscosity (basaltic):**

- ~50% $\mathrm{SiO_2}$
- Flows easily
- Volatiles escape gradually
- **Effusive** eruptions
- Broad shield volcanoes, flood basalts
- Moon, Mars, Io

**High viscosity (silicic):**

- >65% $\mathrm{SiO_2}$
- Traps volatiles
- Pressure builds until failure
- **Explosive** eruptions
- Steep stratovolcanoes
- Primarily Earth

# Olympus Mons: Largest Volcano in the Solar System


Credit: NASA/JPL/USGS

- Shield volcano on Mars
- Base diameter: ~600 km
- Height: ~21.9 km above surroundings
- 2.5× Everest above sea level
- Basal escarpment up to 6 km high

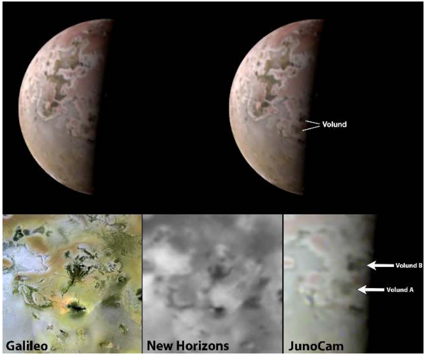
No plate tectonics on Mars → hotspot stays fixed → lava piles up for billions of years in one location.

# Lunar Maria: Flood Basalts

- Vast basalt plains filling ancient impact basins on the nearside
- Erupted between 3.9 and 3.1 Ga (radiometric ages from Apollo samples)
- Cover ~16% of the lunar surface
- Visible from Earth as the dark patches forming the “face” of the Moon
- Record a period of residual internal heating, now extinct

On Earth, plate motion carries the crust over hotspots → chains of smaller volcanoes (e.g., Hawaiian Islands) rather than single massive edifices.

# Venus & Io: Active Volcanic Worlds



Credit: NASA/JPL-Caltech

**Venus:**

- Lava plains cover ~80% of surface
- >1600 volcanic centres
- Stagnant lid; possible episodic overturns

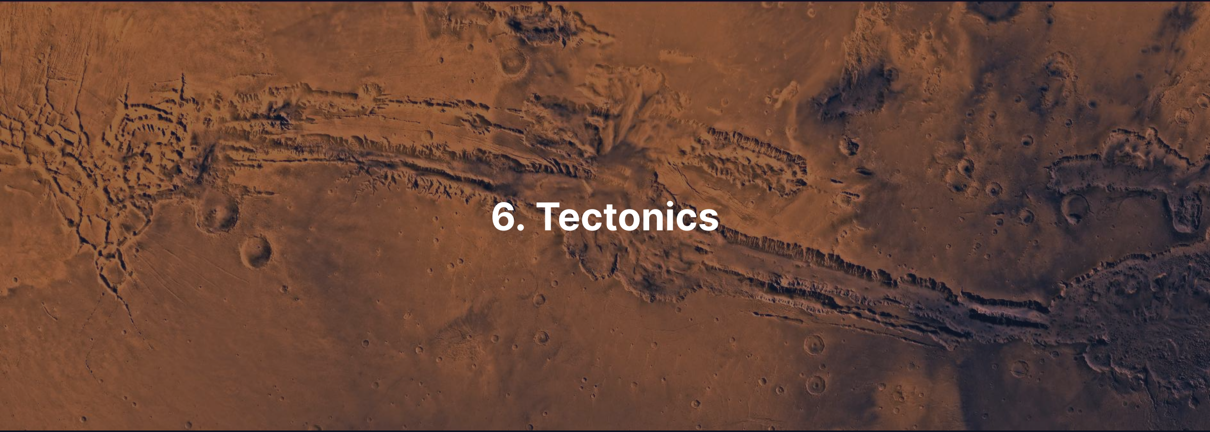
**Io:**

- Most volcanically active body
- ~400 active volcanic centres
- Powered by tidal heating (Laplace resonance)
- Surface age <1 Myr

# Volcanism Across the Solar System

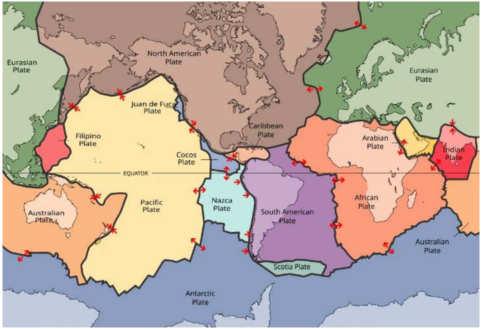
| Body | Style | Driving mechanism | Examples |
|---|---|---|---|
| Earth | Effusive + explosive | Internal heat + recycling | Hawaii, mid-ocean ridges |
| Moon | Flood basalts (extinct) | Residual heat (3.9–3.1 Ga) | Maria |
| Mars | Shield volcanoes | Mantle plumes, no plates | Olympus Mons, Tharsis |
| Venus | Lava plains + shields | Internal heat, stagnant lid | Maat Mons |
| Io | Ultramafic flows | Tidal heating | Loki Patera, Pele |

# 6. Tectonics



Valles Marineris, Mars. Credit: NASA/JPL/USGS

# Plate Tectonics: Earth's Unique Regime




Credit: Wikimedia, CC BY-SA 3.0

Earth is the **only** body with active plate tectonics:

- ~15 rigid plates, moving at 1–10 cm $yr^{-1}$
- **Divergent:** mid-ocean ridges (new crust)
- **Convergent:** subduction zones (recycling)
- **Transform:** horizontal sliding

Enables the **carbonate-silicate cycle**: Carbon recycling → long-term climate regulation.

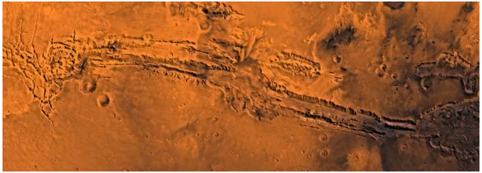
# Stagnant-Lid Convection

- All other terrestrial bodies: **single rigid lithospheric lid**
- Mantle convects beneath, but lid does not participate
- Heat escapes by conduction through the lid and occasional volcanism
- Lid thickens over time as interior cools → volcanism eventually shuts down

**Why is Earth unique?**

- Viscosity contrast at lid base: ~ $10^{10}$
- Breaking the lid requires water weakening + self-sustaining damage
- Likely a combination of size, water content, and thermal history

# Mars: Tharsis and Valles Marineris



Credit: NASA/JPL/USGS

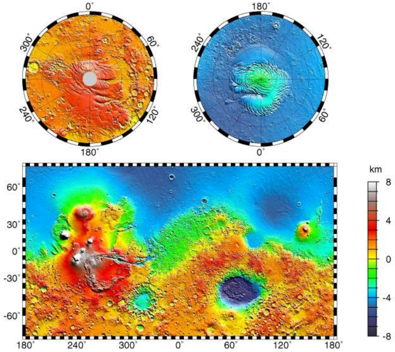
**Tharsis bulge:**

- Volcanic plateau ~5000 km across, ~10 km high
- Likely uplifted by a mantle plume

**Valles Marineris:**

- Largest canyon system: ~4000 km long
- Up to 7 km deep, 200 km wide
- Extensional rifting from Tharsis
- Lisbon to Moscow in scale

# Mars Topography: The Hemispheric Dichotomy




Credit: NASA/JPL/GSFC (MOLA)

~6 km elevation difference between southern highlands and northern lowlands. Tharsis bulge with 4 giant volcanoes; Hellas basin (–8.2 km); Valles Marineris at equator.

# Venus & Mercury: Contrasting Tectonic Styles

**Venus, episodic resurfacing:**

- Uniform crater density → mean age ~300–700 Myr
- Hypothesis: periodic catastrophic lid foundering
- Entire surface resurfaced volcanically in a short interval
- Between episodes: tectonically quiet

**Mercury, global contraction:**

- Large iron core cooled and solidified over time
- Planet's radius shrank by ~7 km
- Produced **lobate scarps**: thrust faults up to several hundred km long, 1–3 km high
- Mapped by Mariner 10 and MESSENGER

# 7. Erosion & Weathering

Fluvial channels, Mars. Credit: NASA/JPL-Caltech/U. Arizona

# Aeolian (Wind) Processes

Wind erosion and deposition require a sufficiently dense atmosphere:

- **Mars:** Extensive dune fields in craters and polar regions
  Global dust storms loft particles to 30 km altitude
  Active dune migration observed by Curiosity
- **Titan:** Vast equatorial dune fields of organic particles (tholins)
  Longitudinal dunes up to 150 m tall, hundreds of km long
- **Venus:** Dense atmosphere but slow surface winds (~1 $m\,s^{-1}$)
  Even slow winds can mobilise fine particles in thick atmosphere

# Fluvial (Water) Processes

Credit: NASA/JPL-Caltech/U. Arizona

**Mars valley networks:**

- Noachian-aged (>3.7 Ga)
- Dendritic drainage patterns resembling terrestrial rivers
- Evidence for sustained liquid water flow

**Outflow channels:**

- Hundreds of km long
- Catastrophic groundwater releases

**Titan:** Active methane rivers; Huygens imaged rounded ice pebbles in a dry riverbed (2005).

# Glacial & Chemical Weathering

**Glacial processes:**

- Earth: ice sheets covered ~30% of land at Last Glacial Maximum (~20 ka)
- Mars: polar $\mathrm{CO_2}$ and $\mathrm{H_2O}$ ice caps; mid-latitude features resemble rock glaciers

**Chemical weathering:**

- Earth: silicate weathering by carbonic acid is the critical carbon sink
- Mars: orbital spectroscopy (CRISM, OMEGA) detected hydrated minerals: phyllosilicates, sulfates, carbonates
- Venus: high surface $T$ (~735 K) drives surface–atmosphere reactions

# Remote Sensing of Surfaces: Overview

We can visit only a few sites. Most surface knowledge comes from orbital remote sensing.

Reflectance spectroscopy (UV–IR) identifies minerals via absorption bands

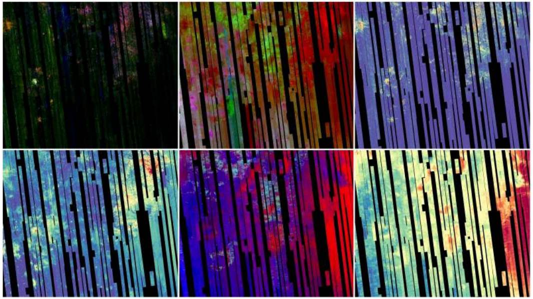
Radar imaging penetrates clouds and regolith, measures surface roughness

Laser altimetry maps topography at sub-metre vertical precision

Gravity field mapping constrains subsurface density anomalies

Each technique probes a different length scale and physical property; together they build a three-dimensional picture of the surface and shallow interior.

# Reflectance Spectroscopy: Mineral Mapping



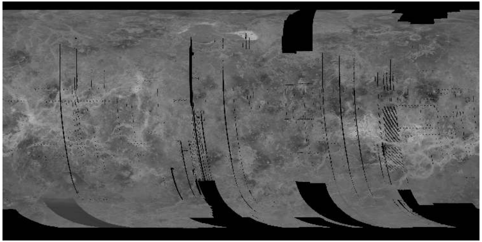
Credit: NASA/JPL-Caltech/JHU APL

CRISM on Mars Reconnaissance Orbiter:

- Maps minerals from near-infrared absorption features
- Fe/Mg phyllosilicates (clays)
- Olivine, carbonates
- Formed by aqueous alteration of basalt

Strongest mineralogical evidence for past liquid water on Mars.

# Radar Imaging: Seeing Through Clouds



Credit: NASA/JPL Magellan Cycle 1

Active microwave imaging is insensitive to clouds and darkness.

**Magellan at Venus (1990–94):**

- SAR mapped 98% of the surface at ~100 m resolution
- Revealed coronae, tesserae, lava flows, young cratered surface

**Cassini at Titan (2004–17):**

- Ku-band radar pierced the organic haze
- Mapped hydrocarbon lakes and seas

Radar is the only way to image surfaces shrouded by thick atmospheres.

# Laser Altimetry and Gravity Mapping

**Laser altimetry (MOLA, LOLA, BELA):**

- Time-of-flight of laser pulses from orbit
- Vertical precision ~1 m over global baselines
- MOLA produced the canonical Mars topographic map (used throughout this lecture)

**Gravity field mapping (GRAIL at the Moon, GRACE/GOCE at Earth):**

- Spacecraft-to-spacecraft tracking measures tiny accelerations
- Gravity anomalies reveal buried mass: dense cores, thin crust, basins, mantle plumes
- Combined with topography, constrains crustal thickness and compensation state

Altimetry and gravity together tell us what the interior **looks like** from above.

# Regolith & Space Weathering

Credit: NASA/Apollo 11

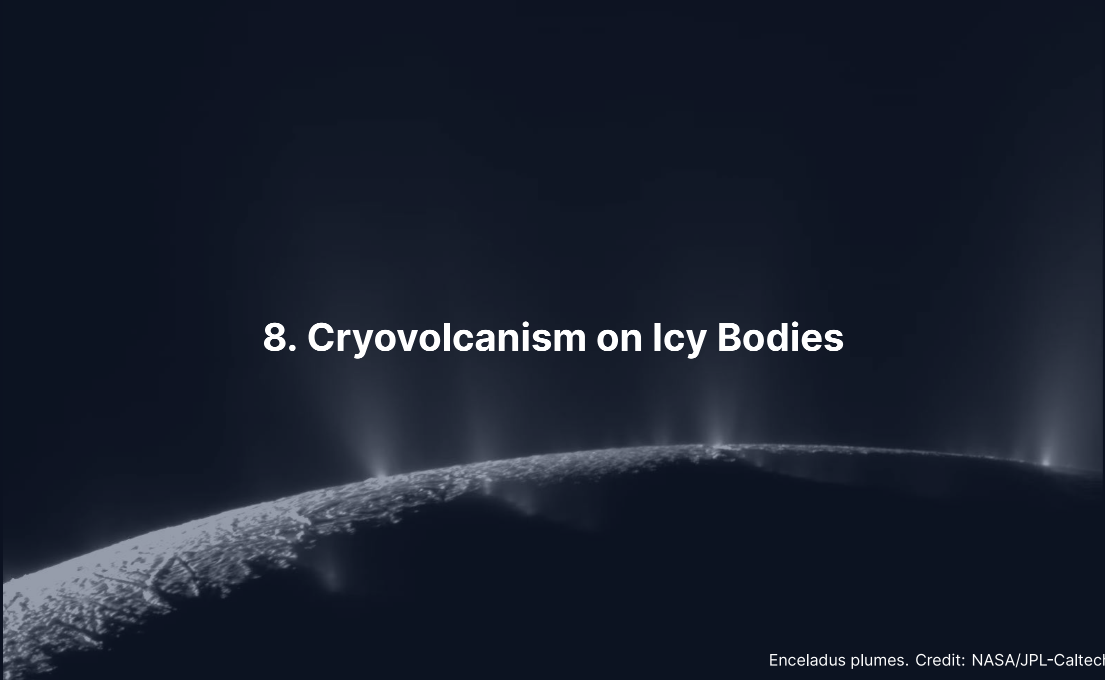
**Regolith:**

- Loose, fragmented surface layer on airless bodies
- Produced by billions of years of impact gardening
- Lunar regolith: 5–15 m thick

**Space weathering:**

- Solar wind ion implantation
- Micrometeorite melting → nanophase iron
- Surfaces become **darker and redder**
- Fresh craters appear bright (e.g., Tycho)

# 8. Cryovolcanism on Icy Bodies



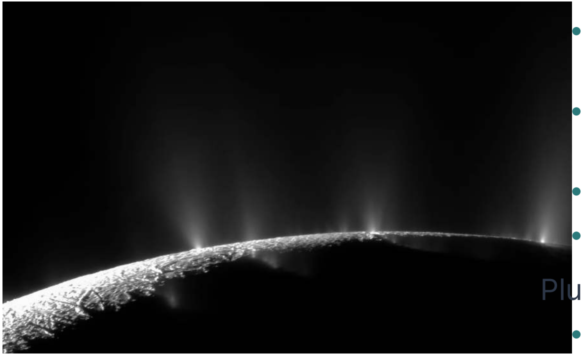
Enceladus plumes. Credit: NASA/JPL-Caltech/SSI

# Cryovolcanism: Ice World Volcanism

In the outer solar system, "volcanic" eruptions involve volatiles rather than silicate melts:

- Erupted materials: liquid water, ammonia–water mixtures, methane, nitrogen
- Energy source: **tidal heating** maintains subsurface oceans beneath icy shells
- "Magma" = low-viscosity volatile liquids
- Analogous to silicate volcanism, but at temperatures of 100–270 K

# Enceladus: Tiger Stripes and Geysers



Credit: NASA/JPL-Caltech/SSI (Cassini)

- Radius only 252 km
- Water ice jets from 4 parallel “tiger stripe” fractures
- Sourced from **global subsurface ocean**
- Ocean in contact with rocky core
- Thermal emission: ~15 GW

Plume composition:

- $H_2$, silica nanoparticles, complex organics
- Consistent with hydrothermal vents

# Europa: A Hidden Ocean

- Jupiter's moon, $R \approx 1561$ km
- Global ocean ~100 km deep; ice shell ~10–30 km thick
- Maintained by tidal heating (Laplace resonance with Io and Ganymede)

Surface features:

- **Lineae:** cracks filled by upwelling ocean water or warm ice
- **Chaos terrain:** broken, rotated, refrozen ice blocks
- **Very few craters:** surface age ~40–90 Myr

**Europa Clipper** (launched 2024): dozens of close flybys to characterise the ice shell, ocean, and habitability.

## Triton & the Ocean Worlds

**Triton** (Neptune's largest moon):

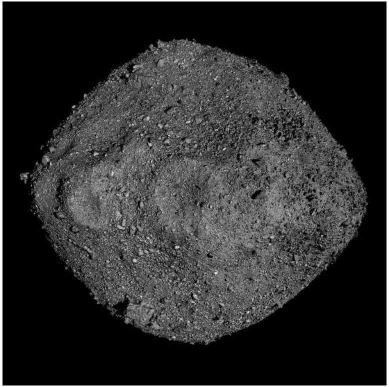
- Nitrogen geysers observed by Voyager 2 (1989)
- Plumes of $\mathrm{N}_2$ gas and dark dust, rising ~8 km
- Very young surface; retrograde orbit (captured KBO)
- Possible subsurface ocean

Enceladus, Europa, and Triton demonstrate that liquid water can exist far beyond the classical habitable zone, maintained by tidal heating rather than stellar radiation.

# 9. Recent Advances

Mars surface. Credit: NASA/JPL-Caltech/MSSS

# OSIRIS-REx: Rubble-Pile Asteroids



Credit: NASA/GSFC/University of Arizona

NASA's OSIRIS-REx returned ~70 g of asteroid **Bennu** in September 2023.

- ~500 m rubble-pile asteroid
- Near-Earth orbit, carbonaceous (B-type), water-rich
- Hydrated clays, carbonates, phosphates
- Amino acids and nucleobases
- Extremely porous: a "loose pile of rocks"
- Insight into early solar system aqueous alteration

Source: Lauretta et al. (2019, 2024)

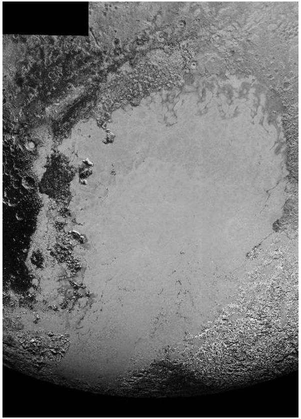
# Hayabusa2: Sampling Ryugu

JAXA's Hayabusa2 delivered 5.4 g of asteroid Ryugu to Earth in December 2020.

- C-type carbonaceous asteroid, 900 m diameter
- First sample of a water-bearing asteroid returned to Earth
- Contains pristine organic molecules: amino acids, sugars, uracil
- Evidence for aqueous alteration on the parent body
- Noble gas and rare-earth abundances constrain its formation location
- Complements OSIRIS-REx by sampling a different asteroid

Source: Watanabe et al. (2019); Yada et al. (2022)

# Pluto: A Surprisingly Active World



Credit: NASA/JHUAPL/SwRI (New Horizons 2015)

New Horizons (2015) revealed Pluto's geological diversity.

- **Sputnik Planitia:** 1000 km nitrogen ice glacier
- Cellular convective patterns: active $N_2$ ice convection today
- Surface age ≲ 10 Myr
- Mountains of water ice up to 3 km tall
- Possible subsurface liquid water ocean

Source: Stern et al. (2015); Moore et al. (2016)

# Perseverance Rover at Jezero Crater

- Operating in Jezero crater since February 2021
- Crater floor: igneous rock (olivine-bearing cumulates)
- Subsequently altered by liquid water
- >20 sample tubes cached for Mars Sample Return
- Joint NASA/ESA mission: first laboratory analysis of Martian rocks
- Will address past habitability and possible biosignatures

Jezero was chosen because it is an ancient lake crater with a preserved delta deposit, maximising the chance of finding evidence for past life.

# DART: Planetary Defence Experiment

- **Double Asteroid Redirection Test** (September 2022)
- Kinetic impactor deliberately crashed into moonlet **Dimorphos**
- Changed orbital period around Didymos by **33 minutes**
- First successful planetary defence experiment
- Confirmed kinetic impact as a viable deflection strategy
- Impact ejecta revealed properties of rubble-pile asteroid surfaces

DART demonstrated that we can deflect a potentially hazardous asteroid using existing technology.

Source: Daly et al. (2023), *Nature*; Thomas et al. (2023)

# Ongoing Exploration: Io and Venus

**Io monitoring:**

- Juno extended mission: close Io flybys in 2023–2024
- Discovery of previously unknown eruption sites
- Constraints on spatial distribution of tidal heat flow
- Ground-based adaptive optics complement spacecraft data

**Upcoming Venus missions:**

- VERITAS and EnVision (planned for early 2030s)
- First high-resolution surface radar data since Magellan (1990–1994)
- Will test whether Venus has recent or ongoing volcanic activity

# Summary

1. Surfaces record the competition between **endogenic** (volcanism, tectonics) and **exogenic** (impacts, erosion) processes
2. Impact cratering is universal; three stages: contact, excavation, modification
3. Crater scaling law: $D \sim (E/\rho g)^{1/4}$ (fourth-root dependence on energy)
4. Crater counting + Apollo calibration → absolute surface ages
5. Volcanism style depends on magma viscosity; no plate tectonics → giant volcanoes
6. Earth is unique in having plate tectonics; other bodies have stagnant lids
7. Fluvial features on Mars: strong evidence for past liquid water
8. Cryovolcanism on Enceladus, Europa, Triton: liquid water beyond the habitable zone

# Questions?

Next (after mid-term exam): **Lecture 8. Planetary Interiors: Structure, Composition, & Dynamics**

Lecture notes: `formingworlds.github.io/IntroductionToPlanetaryScience`